Unit - 7
Thermodynamics

Evaluation Yourself

1.

Solution :

Given

$\Delta H_f^0\ CO_2 = -393.5\ kJ\ mol^{-1}$

$\Delta H_f^0\ CO = -111.31\ kJ\ mol^{-1}$

$\Delta H_f^0\ (H_2O) = -242\ kJ\ mol^{-1}$

$CO_2(g) + H_2(g) \rightarrow CO(g) + H_2O(g)$

$\Delta H_r^0 = ?$

$\Delta H_r^0 = \Sigma\ (\Delta H_f^0)_{products} - \Sigma\ (\Delta H_f^0)_{reactants}$

$\Delta H_r^0 = [\Delta H_f^0\ (CO) + \Delta H_f^0\ (H_2O)] - [\Delta H_f^0\ (CO_2) + \Delta H_f^0\ (H_2)]$

$\Delta H_r^0 = [-111.31 + (-242)] - [-393.5 + (0)]$

$\Delta H_r^0 = [-353.31] + 393.5$

$\Delta H_r^0 = 40.19$

$\Delta H_r^0 = +40.19\ kJ\ mol^{-1}$

2.

Solution :

Given :

number of moles of water n

$= \frac{180\ g}{18\ g\ mol^{-1}} = 10\ mol$

molar heat capacity of water

$C_P = 75.3\ J\ K^{-1}\ mol^{-1}$

$T_2 = 100^0\ C = 373\ K$

$T_1 = 25^0\ C = 298\ K$

$\Delta H = ?$

$\Delta H = nC_P\ (T_2 - T_1)$

$\Delta H = 10\ mol \times 75.3\ J\ mol^{-1}\ K^{-1} \times (373 - 298)\ K$

$\Delta H = 56475\ J$

$\Delta H = 56.475\ kJ$

3.

$$C_6H_6\ (l) + \frac{7}{2} O_2(g) \rightarrow 6\ CO_2\ (g) + 3\ H_2O\ (l)$$

ΔU at $25^0\ C = -3268.12\ kJ$

Solution :

Given

$T = 25^0\ C = 298\ K$;

$\Delta U = -3268.12\ kJ\ mol^{-1}$

$\Delta H = ?$

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta H = \Delta U + (n_p - n_r)\ RT$

$$\Delta H = -3268.12 + \left(6 - \frac{15}{2}\right) \times 8.314 \times 10^{-3}\ x\ 298$$

$\Delta H = -3268.12 - (1.5 \times 8.314 \times 10^{-3} \times 298)$

$\Delta H = -3268.12 - 3.72$

$\Delta H = -3271.84\ kJ\ mol^{-1}$

4.

Solution :

Given :

$Mg(S) + Br_2(1) \rightarrow$

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